

Candidate's Name:

Signature:

Index Number							

UACE
Chemistry Paper 1
Theory
S.5 2025
3hours



NELSON MANDELA SECONDARY SCHOOL
End of year exams
S.5 Chemistry Theory paper 1

Paper 1
Theory
3 Hours

INSTRUCTIONS

- ✧ *This paper consists of six examination items in two sections A and B.*
- ✧ *Write your responses in the gaps left for section A and use the answer booklets given for responses in section B*
- ✧ *Respond to both the items in section A.*
- ✧ *For section B, Items 3 and 4 are compulsory.*
- ✧ *Respond to three items in section B including the two compulsory items 3 and 4*
- ✧ *Both the two last items, 5 and 6 in section B carry equal scores. No any extra item attempted will be scored.*

(40 Scores)

A fire extinguisher manufacturing industry uses hydrocarbons as the extinguisher gas. When aimed at fire, the hydrocarbons burn to produce carbon dioxide gas which extinguishes fire. However, the liquid substance formed on burning of the hydrocarbons becomes a nuisance to the extinguishing process as it tends to boost the fire propagation. On analysis, the hydrocarbon that produces less amount of this liquid is the more suitable for this activity. Hydrocarbons J of molar mass 56 and K of molar mass 58 were analyzed to have similar carbon content but its percentage proportion by mass in the two two hydrocarbons is 85.7% and 82.8% respectively. The industrial director has totally failed to understand what the analysts tell him and so they seek help from you.

- a) Help the analysts to
- i) Deduce the IUPAC identity of each hydrocarbon.

K.

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04scores

J. 02scores

K. 02scores

- b) i) Demonstrate to the director using chemical equations how each of the compounds J and K acts to extinguish the fire.

J. 02scores

K. 02scores

- i) Basing on the stoichiometry of combustion of the two hydrocarbons, which one would more suitably be used as the extinguisher gas? (State a reason)

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..... 02scores

- c) Compare the volume of carbon dioxide that is produced by each gas if a full cylinder that accommodates 500g of gas is separately used to put off fire until it is empty at s.t.p. (1mole of a gas occupies 22.4dm³ at s.t.p.)

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..... 03scores

Item 2

A certain group of boys in a school Science Club consisting of Paul Nathan and other members came across a certain greyish white metal which they suspected to be aluminium. They wanted to prove whether the metal they were expecting was right. Paul suggested heating it in steam. They heated approximately 0.52g of the metal piece in steam which produced a gas they did not know. The gas was collected, cooled and measured at room temperature. Nathan seemed unsatisfied and he suggested addition of dilute nitric acid to the solid residue that remained after heating the metal in steam. To 2cm³ of the resultant solution was added sodium hydroxide drop-wise until in excess, and to an other 2cm³ was added ammonia solution drop-wise until in excess. The remaining solution was evaporated to complete dryness. 0.72g sample of the dry solid obtained was heated strongly until there was no further change. The students wanted to record the mass of the solid residue that remained after heating but they lacked a weighing scale to determine the accurate mass and they need your help.

Hint: *1mole of a gas occupies 22.4dm³, Al=27, O=16, H=1*

Task

As a chemistry student, (supposing the metal they expected was right)

- a) i) State what the boys observed and demonstrate what chemically took place when what Paul suggested was done.

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..... 03scores

ii) Determine the volume of the volume of the gas that was collected and measured at r.t.p. (1mole of a gas occupies 24dm³ at r.t.p.)

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..... 02scores

b) State what was observed and ionically demonstrate what chemically took place when Nathan's suggestion was done after steam reaction.

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..... 03score

c) State what was observed and write ionic equation(s) where necessary to demonstrate the reactions that took place with excess sodium hydroxide and ammonia solutions.

i) With sodium hydroxide.

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..... 03scores

ii) With ammonia solution.

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..... 02scores

d) i) Show what the boys would observe when the dry substance obtained after evaporation was heated strongly.

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..... 03scores

iii) Help the boys to determine the mass of the solid residue that remained after strong heating without the weighing scale.

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..... 02scores

SECTION B

Item 3

A alcohol production plant manufactures two isomeric alcohols **L** and **M**. The two alcohols are supplied to scientific research institutions and laboratories. A chloroalkane **J** molar mass **78.5** containing **45.86%** and **8.92%** by mass of carbon and hydrogen respectively is used as a raw material in the first reaction vessel where it is converted to an unsaturated hydrocarbon **K** with double bonds at one terminal end of its carbon skeleton. Hydrocarbon **K** is directed into the second vessel containing an acid from which an alcohol **L** is produced. Here, some alcohol is packed while the remaining is passed through the third vessel containing an biological enzyme **isomerase** so as to obtain its alcoholic isomer **M**. To avoid wastage, excess alcohol **L** is **recycled back** to form hydrocarbon **K**. This is done through two alternative routes **X** or **Y**. In route **X**, **an acid** is used while route **Y** involves the use of a **solid ionic compound** obtained from decomposition of a metal carbonate by strong heat. The alcohol **M** is used as a **fuel** in the machine engines of the factory because it produces much heat energy on **combustion**. The factory is designed in such a way that the water produced from combustion of the fuel alcohol is diverted into water jackets to **cool** the engines. To avoid pollution, the **gaseous** product is collected into cylinders for fire security and awareness. Each fuel tank of the machine consumes **55g** of the fuel alcohol **each hour**.

Task

- a) Determine the molecular, and structural formula and the IUPAC identity of chemical **J**.
03scores
- b) Explain with names of reagents, conditions and equations (also show the mechanisms of reactions) the reaction that takes place in,
- i) The first reaction vessel leading to conversion of **J** to **K**. 03scores
- ii) The second reaction vessel when **K** is converted to **L** 03scores
- iii) The recycle process when excess **L** is converted back to **K** using route **X**.
03scores
- c) Name the reagent used in route **Y** and show how it is chemically obtained according to the scenario.
02scores
- d) i) Write the structural formula and name alcohol **M**. 01score
- iii) Chemically illustrate how alcohol **M** burns in the engine. Hence determine the volume of water produced each hour to cool the engine and the volume of the gas collected at s.t.p. (Density of water is 1gcm^{-3} , 1mole of a gas occupies 22.4dm^3)
05scores

Item 4

In Nakitokolo subcounty, there is only one hospital where blood transfusion services are offered to clients. One day, all the blood banks were depleted of blood due to a large number of accident casualties who needed blood transfusion. There were five people who were ready to donate blood and **$1.7\text{dm}^3(1500\text{cm}^3)$** were ably collected from them. However, The chemical tests showed that the blood was very alkaline at **PH 10.6** due to presence of a **basic solution B**, (NH_4OH) which had accumulated as a result of excess muscular activity when running to save the casualties hence, it was not recommendable for use on patients. In the drug shelves, there was a full bottle of an **acid A**, labeled as Propanoic at **P^H4**. A small volume of this acid was required to be added to the blood for complete neutralisation of the alkali **B**. However, it needs a chemist to properly determine the exact volume of the acid to be mixed with the whole blood available. Your help is needed, given that the Basic dissociation constant **K_b** of ammonia is **$1.584 \times 10^{-5}\text{mol dm}^{-3}$** and the dissociation constant of the acid is **0.2%**.

Task

You are tasked to;

- Write down the K_a and K_b expressions for the acid, A and the basic solution, B in blood. 04scores
- Determine the molar concentrations of the acid A and the basic substance B in the blood. 10scores
- Determine the number of moles of ammonia in the 1.7dm³ of blood. 03scores
- Number of moles of the acid that would react with the basic solution in blood and hence the exact volume of this acid that should be added to the blood for complete neutralisation. (The reaction ratio of the acid : base = 1 : 1) 05scores

Item 5. Anitah, a S.5 PCM class student at Nelson Mandela S.S, wants to make a project of making an electronic cell. She has two ionic salts; Potassium Iodide and Potassium Chloride, one of which she has to choose for use as the electrolyte. It is well known to her that the salt with the higher solubility in water or less stable is more suitable for use in this project. However, she is confused of which one of the two. When she consulted the chemistry teacher for advice, she was given the following enthalpy data in table 1 to use for self direction. But she failed to know how to use it and she needs your help.

Enthalpy description	Enthalpy value (KJ/mol)
Formation of potassium iodide	-327.9
Formation of Potassium Chloride	-436.7
Hydration of potassium Iodide	-628
Hydration of potassium Chloride	-685
Sublimation of potassium	89
First ionization of potassium	419
Bond dissociation of Iodine	151
First electron affinity of Iodine	-295
First electron affinity for Chlorine	-349
Bond dissociation energy of Chlorine	242.3

Task

As her fellow chemistry student, Help Anitah to;

- Identify the more suitable electrolyte for use in her project basing on stability. State a reason for your suggested choice. 02scores
- Explain the two factors that affect the solubility of the electrolytes. 03scores
- Determine the lattice energy of each electrolyte she has using a suitable energy diagram. 06scores
- Determine the enthalpy of solution for the each electrolyte. Using born haber cycles. 05scores
- Comment on the results and recommend Anitah on which salt she should use for her project.

Item 6

A chemical manufacturing industry deals in the manufacture of nitric acid and supplies it to different chemical laboratories in Uganda. The production of the acid is based on Haber process where 0.3moles of nitrogen are normally reacted with 0.5moles of hydrogen.



The amount of ammonia produced at equilibrium is normally 0.3moles. The volume of ammonia produced at s.t.p determines the amount of acid that is produced and the degree of dissociation determines the stability of the gas at equilibrium, the equilibrium constant K_c value and the position of the equilibrium. Due to the increased demand of ammonia solution and nitric acid on the market, the rate at which ammonia is produced must be increased. According to the industrial chemical analysts, if the degree of dissociation of ammonia is below 0.2, ie 20%, the reaction conditions must be altered to adjust the equilibrium value, position and the rate of production of ammonia. Among the conditions to be altered are concentration of the reactant gases, pressure and temperature in the reaction vessel. The reaction vessel can accommodate a maximum of 1dm^3 of gaseous products. The industrial analysts also advised to introduce fine powder of iron metal into the reaction vessel as well as constant removal of ammonia from the vessel as promptly as possible. However, the industrial manager and his workers are confused of what the analysts are talking about because it seemed to be beyond their understanding and they seek for your help.

Task

As a chemistry student, Help the manager and his team to

- Determine the volume of ammonia that is normally produced at equilibrium at s.t.p. (02scores)
- Determine the degree of dissociation of ammonia and the equilibrium constant K_c . From your results, draw a conclusion on whether to alter the conditions or not according to the advise of the analysts. 05score
- If the conditions were to be altered; Explain with genuine reasons how the alteration of each condition would affect the production process. 06scores
- Briefly describe the process of production of nitric acid using the ammonia obtained from this industrial process. 03scores
- Explain the effect of introducing fine iron metal powder into the reaction vessel and the prompt removal of ammonia from the vessel. 02scores

*****Merry Christmas and a happy new year*****